

MAINTENANCE WORK ORDER

Work Order No. : WO-4476
Date Issued : 22 August 2024
Priority : MEDIUM
Status : CLOSED ? PARTIAL (see Notes)
Department : Mechanical
Location : Bay 3, Zone C
Equipment Tag : HX-11 (Shell-and-Tube Heat Exchanger, Train 2)
Assigned To : Suresh Kumar (Plant Operator), Anil Mehta (I&E support)
Supervised By : Rajiv Sharma, Maintenance Supervisor

1. WORK DESCRIPTION

Investigation of odour noted at HX-11 south shell flange during Night Shift of 19 August 2024, per Shift Handover Log SHL-2024-0819. Odour described as "faint gas smell near south flange, not confirmed as leak."

This work order is linked to WO-4471 (GB-14 investigation) ? the elevated GB-14 reading on 19 August 2024 was attributed in part to a possible HX-11 flange seepage. WO-4476 is to confirm or rule out the HX-11 source.

2. WORK PERFORMED

Date: 23 August 2024

Step 1 ? Soap bubble test on HX-11 south flange joints (Flange #5, #6, #7, #8):
 Flange #7: Small bubble formation observed at 3 o'clock position. Minor seepage.
 Flanges #5, #6, #8: No bubbles. No leak.

Step 2 ? Portable gas reading at HX-11 Flange #7: 4% LEL methane. Source confirmed.

Step 3 ? Flange #7 bolts: 4 of 8 bolts at reduced torque (measured: 65?72 Nm vs. 110 Nm specification). Two bolts showing corrosion on threads.

3. REMEDIAL ACTION

Bolts retorqued to 110 Nm. Post-retorque soap test: No bubbles. Seepage appears resolved. Portable gas at Flange #7: 0.5% LEL. Acceptable.

? IMPORTANT NOTE: Flange #7 gasket (spiral-wound, 150# ANSI class) should be replaced at next planned shutdown. The gasket shows surface fretting from repeated thermal cycling and the bolt torque loss indicates potential creep. Retorquing is a TEMPORARY FIX only.

Recommended: Schedule gasket replacement at next Bay 3 planned maintenance window (Q4 2024 or Q1 2025 shutdown).

4. CLOSURE STATUS

CLOSED ? PARTIAL. Immediate seepage resolved by retorquing. Gasket replacement required as permanent fix. Planned maintenance team notified verbally by Rajiv Sharma.
Work Order for gasket replacement: NOT RAISED (to be raised at next planned shutdown).

[Post-incident note added 22 January 2026: Gasket replacement was NOT completed. HX-11 Flange #7 gasket failure occurred on 21 January 2026 and was identified as the methane source in WO-4892. The failure to raise a formal follow-up work order for gasket replacement is identified as a contributing factor in RCA-2026-007.]

5. SIGNATURES

Technician	: Suresh Kumar	Date: 23/08/2024
Supervisor	: Rajiv Sharma	Date: 23/08/2024

CROSS-REFERENCE: WO-4471, SHL-2024-0819, WO-4892, RCA-2026-007